Trustworthiness is not one property: accuracy, cross-dataset robustness, explainability, abstention, and deployment cost in single-cell cervical cytology classification

Md. Mostafizur Rahman*

Department of Electrical and Computer Engineering, San Francisco Bay University, 161 Mission Falls Lane, Fremont, California 94539, United States

Abstract. Purpose: Lightweight vision transformers are proposed for automated cervical cytology screening, but "trustworthy AI" is usually evaluated as one property that a pipeline has or lacks. We tested whether accuracy, cross-dataset robustness, explainability, selective prediction, and deployment cost agree with one another on one single-cell cervical cytology classification task, using a common model registry but different backbones for different properties. **Approach:** We benchmarked 12 convolutional, full-size transformer, and efficient transformer models on SIPaKMeD (five classes) and evaluated all 12 zero-shot on three external cytology datasets by mapping their outputs to a binary normal/abnormal label: Herlev (single-cell crops, the primary external test) and Mendeley LBC and APCEM (multi-cell fields of view, the latter algorithmically enhanced, treated as content-shift stress tests), so the external evaluation combines differences in institution, scale, content, preprocessing, and annotation. On a standard vision transformer, we compared three explainability methods against deletion/insertion faithfulness and expert-annotated nucleus boundaries, tested a combined confidence and conformal-set-size abstention rule, and measured parameters, floating-point operations (FLOPs), central processing unit (CPU) latency, and explanation cost. **Results:** The two smallest models (about 2.1M parameters) were the two most accurate in distribution in the primary run and ranked second and third by three-seed mean, on a cell-level split in which most test cells share a source image with training cells, which could inflate in-distribution scores; a cluster-grouped re-run changed balanced accuracy by a median of -0.9 points across the 12 models (range -14.6 to $+1.1$, the extreme being a training failure), comparable to seed variation. In-distribution accuracy did not predict robustness consistently: the in-distribution leader ranked 10th of 12 on both Herlev and Mendeley LBC in the primary run, and on APCEM (multi-cell fields) 9 of 12 models fell below chance on the area under the receiver operating characteristic curve. Rank correlations between three-seed mean external accuracy and in-distribution accuracy, parameter count, and FLOPs were weak to moderate and changed sign across datasets, and model rankings changed across training seeds and datasets. Relevancy propagation was the most faithful explainability method on all three metrics on two backbones, with paired cluster-bootstrap intervals excluding zero in all 12 comparisons over 596 test cells (nucleus concentration 3.8 times chance versus 1.3 for attention rollout on DeiT-Tiny), although concentration on the nucleus does not establish clinical validity. Combining confidence with conformal set size helped only under severe shift (risk-coverage area under the curve 0.286 versus 0.324) and hurt in distribution and under mild shift. FLOPs did not predict CPU latency, and an explanation cost about 3.3 times a bare prediction. **Conclusions:** Accuracy, robustness, explainability, abstention, and efficiency resolved differently across experiments on the same task, so each needs its own test under the conditions of intended deployment; whether a joint test on a single backbone would change these conclusions was not tested.

Keywords: cervical cytology, vision transformers, cross-dataset generalization, explainability, selective prediction, model efficiency.

*Corresponding author: Md. Mostafizur Rahman, csemrm@gmail.com; ORCID: 0009-0007-2022-8677

1 Introduction

Cervical cancer remains a significant cause of cancer morbidity and mortality worldwide[1] and is substantially more curable when detected at a localized stage, yet access to screening and specialist cytology capacity are limited in many settings.

"Trustworthy AI" is usually evaluated as if it were a single property. In practice a screening pipeline makes at least five separable claims: (i) a lightweight model gives up little accuracy; (ii) its accuracy transfers across institutions, scanners, and staining protocols; (iii) its explanations are faithful; (iv) its confidence supports a reliable abstention rule; and (v) its efficiency claim holds under real measurement. A reader might reasonably assume that these resolve alike, so that testing one is enough. We test them side by side on one cervical cytology classification task and ask whether they do. The experiments share a dataset family and a 12-model registry, but, as Sec. 6 states, they do not all use the same backbone: the benchmark covers 12 models, the explainability and abstention experiments use DeiT-Tiny, and the degradation experiment uses EfficientViT-M0.

We make four contributions.

1. **Efficiency–accuracy benchmark.** A 12-model benchmark on SIPaKMeD spanning convolutional neural network (CNN), full-size vision transformer, and efficient vision transformer families, in which the two most parameter-efficient models are also the most accurate in distribution.

2. **Domain-shift evaluation.** A zero-shot evaluation of all 12 models on three independently sourced external datasets (Herlev, Mendeley LBC, APCEM), together with a three-seed sensitivity check, showing that in-distribution accuracy, parameter count, floating-point operations (FLOPs), and architecture family did not predict which model fails safely consistently across datasets (the direction of the association changed), and that even apparent family-level patterns did not survive a third dataset or a change of seed.

3. **Explainability and abstention.** A comparison of relevancy propagation, attention rollout, and Grad-CAM against behavioral faithfulness and expert-annotated nucleus boundaries, replicated on a second backbone, and a test of whether combining temperature-scaled confidence with conformal prediction-set size improves selective prediction.

4. **Measured deployment cost.** A measurement of parameters, FLOPs, central processing unit (CPU) latency, and explanation cost, showing that FLOPs do not predict latency and that returning an explanation costs roughly 3.3 times a bare prediction.

The properties resolve differently: explainability method choice resolves cleanly, abstention composition resolves conditionally on shift severity, and robustness and efficiency resolve only under

direct measurement. This divergence across experiments, rather than any single result, is the central observation. Because the backbones differ between experiments, the study shows that these properties are not interchangeable, but it does not show that a joint test on a single backbone is necessary to see this.

2 Related Work

2.1 Efficient Vision Transformers

Efficiency-oriented transformer variants fall into several loose lineages. Hybrid local–global designs (MobileViT[2], MobileViTv2[3]) interleave convolutional stages with transformer blocks. Distillation-oriented designs (DeiT[4]) keep a standard ViT backbone[5] but reach competitive accuracy with far less pretraining data. Hardware-aware designs (LeViT[6], TinyViT[7]) reintroduce convolutional stages to improve real-device latency rather than FLOP count alone. EfficientViT-M0 and EfficientViT-B0, the two smallest models in our registry, are often assumed to share a lineage because `timm` groups their tags; verified directly, they do not. EfficientViT-M0 uses cascaded group attention[8], whereas EfficientViT-B0 belongs to an unrelated lineage built on multi-scale linear attention for dense prediction[9]. These design papers do not evaluate cross-institution domain shift as part of their efficiency claims, so efficiency and robustness are treated as unrelated properties. Parameter count, FLOPs, and measured speed are also known to disagree as cost indicators in mobile and efficient network design,[2,10,11] although we found no study of this disagreement for cytology models.

2.2 Domain Shift and Model Selection in Medical Imaging

A general medical-imaging literature documents that models selected on held-out in-distribution data can degrade under real distribution shift across scanners, sites, and populations,[12,13] motivating external validation. Benchmarks of real-world distribution shift,[14] work on shortcut learning,[15] and work on underspecification[16] show why in-distribution accuracy can be a poor guide: models that perform equivalently on held-out data can behave differently once the data change. Predictive uncertainty also degrades under shift.[17] In cervical cytology, recent studies have examined classifier behavior on real-world and multi-scanner data,[18,19] and external validation across independent cervical cytology datasets has been described as rare.[20] That literature typically evaluates one or a few architectures. We instead sweep CNN, full-size transformer, and efficient transformer families at matched task and data scale and ask whether any measurable static model property predicts which model degrades least.

2.3 Data Splitting and Source-Image Overlap

Splitting at the level of images, tiles, or records when several samples come from the same subject can inflate accuracy relative to subject-level splitting, as described in digital pathology,[21] clinical machine learning,[22] and brain MRI.[23] For SIPaKMeD, Albzour and Lam note that the cropped-cell filenames encode the parent cluster image and that parent-level grouping was not enforced in their cross-validation, and they treat this as a limitation without measuring it.[20] We measure the overlap under the split used here (Sec. 3.3.1) but do not estimate its effect on accuracy.

2.4 Transformer Explainability and Faithfulness

Attention rollout[24] chains attention weights across layers as a relevance signal, a premise contested in the literature on whether attention is an explanation.[25,26] Relevancy propagation[27] weights each block's attention by its gradient with respect to the target class, making the map class-discriminative. Grad-CAM[28] was designed for convolutional feature maps and is commonly adapted to transformers by targeting a block's token activations. Deletion/insertion faithfulness[29] measures the effect of removing or revealing the regions an explanation marks as important on the model's own prediction. Evaluating localization against SIPaKMeD's expert-marked nucleus polygons avoids compounding the evaluation with a second model's errors. Prior studies have evaluated explanation methods for medical vision transformers, including Grad-CAM and attention-based methods;[30–32] we did not find prior work in our search that scores explanations against expert nucleus masks in cervical cytology.

2.5 Selective Prediction and Calibration

Temperature scaling[33] calibrates confidence on a held-out split. Split conformal prediction gives a distribution-free coverage guarantee from a single calibration split,[34] and its prediction-set size is a model-agnostic but coarse uncertainty proxy. MC Dropout[35], deep ensembles[36], and evidential deep learning[37] are alternative uncertainty signals that we do not evaluate here. Conformal prediction has been applied to cervical atypia classification,[38] and calibration with selective prediction has been studied for Pap smear classification.[39] In our search we did not find work that combines confidence with conformal set size for abstention under cross-dataset shift in cytology, which is the question we examine.

2.6 Adjacent Cervical and Clinical-AI Work

Comparisons of many convolutional and transformer architectures on cytology images,[40] ensembles that include EfficientViT on SIPaKMeD and Herlev,[41] and systematic evaluations of vision transformers for cervical classification[42] exist; among them we did not identify an evaluation of

zero-shot transfer across independent datasets, judging from their titles and abstracts. Adjacent work includes hybrid deep feature fusion for cervical cell classification[43] and a stacking-ensemble approach with explainable AI for cervical cancer diagnosis,[44] which we differ from by evaluating explainability against expert-annotated localization and behavioral faithfulness and by jointly evaluating efficiency and real cross-institution shift. A lightweight transformer with cross-scale attention and explainability for plant disease diagnosis[45] motivates our treatment of efficiency and explainability as joint design goals, and two further papers address explainable clinical AI in other domains[46,47].

These statements reflect a non-exhaustive literature search conducted in October 2026 (Crossref, arXiv, and web search); the absence of a prior study in our search does not establish that none exists.

3 Materials and Methods

3.1 Datasets

All data are from publicly available, previously published image datasets, and no new patient data were collected. All models were trained on SIPaKMeD[48]: 4,049 native-resolution single-cell crops derived from the dataset's own cytoplasm-polygon annotations, in five cytomorphology classes. External evaluation used Herlev[49], Mendeley LBC[50], and the APCEM-Enhanced Cervical Cytology Pap Smear Image Dataset[51] (4,253 images in the six-class Bethesda taxonomy, with dataset class labels NILM, ASC-US, ASC-H, LSIL, HSIL, and SSC). Because the four datasets use different label taxonomies, external evaluation used one shared binary normal/abnormal label (Table 2). Models were trained on SIPaKMeD's five classes, and each model's five-class softmax output was summed into normal and abnormal probabilities for external evaluation (Sec. 3.4). The external datasets differ from SIPaKMeD and from each other in acquisition pipeline, institution, and label granularity, and the correspondence between their diagnostic categories is an assumption that the dataset documentation does not establish. We therefore describe the external evaluation as cross-dataset generalization under combined acquisition and annotation differences, and we use "domain shift" in this sense throughout; the experiments do not separate acquisition differences from label-definition differences. Each external dataset is an independent test rather than a synthetic corruption of the same source images. The datasets also differ in what an image contains (Table 1): SIPaKMeD and Herlev images are crops of single cells, whereas Mendeley LBC and APCEM images show many cells in a field of view, and the APCEM images were produced by an algorithmic enhancement procedure applied by the dataset authors. All images were resized without cropping to each model's square input size, which changes the aspect ratio of the non-square Mendeley LBC and Herlev images. A classifier trained on single cells is therefore applied to whole

fields in two of the three external evaluations. We treat Herlev, which preserves the single-cell task, as the primary external evaluation, and Mendeley LBC and APCEM as content-shift stress tests rather than as controlled cross-institution tests. The harmonized external sets are imbalanced (Herlev 74% abnormal, APCEM 82% abnormal, Mendeley LBC 36% abnormal, SIPaKMeD 40% abnormal), which is why balanced accuracy is reported.

Table 1: What an image contains in each dataset. Image sizes were measured on a random sample of 60 images per dataset (SIPaKMeD sizes depend on the cytoplasm outline and were not tabulated). All images were resized to the model's square input.

Dataset	Image content	Image size (pixels)	Preprocessing noted
SIPaKMeD	Single-cell crop	Variable (cytoplasm outline plus 4-pixel margin)	Cropped by us
Herlev	Single-cell crop	Median 123 × 108 (range 43–331 by 53–363)	None noted
Mendeley LBC	Field of view with many cells (liquid-based cytology)	All sampled images 2048×1536	None noted
APCEM	Field of view with many cells	All sampled images 224×224	Enhanced by the dataset authors (APCEM procedure)

3.2 Model Registry

Twelve models spanning three families were evaluated: CNN baselines (ResNet-50[52], MobileNetV2[53], EfficientNet-Lite0[54]); full-size vision transformers (ViT-Small[5], Swin-Tiny[55]); and efficient or mobile-oriented vision transformers (DeiT-Tiny[4], LeViT-128[6], TinyViT-5M[7], MobileViT-S[2], MobileViTv2[3], EfficientViT-B0[9], EfficientViT-M0[8]). All `timm` tags were verified to exist with pretrained ImageNet weights, and the two EfficientViT variants' source papers were confirmed against `timm`'s model cards rather than inferred from the shared tag prefix (exact library versions were not recorded; Table 3).

3.3 Training and In-Distribution Evaluation

Models were trained on SIPaKMeD's five classes from ImageNet-pretrained weights with the AdamW optimizer (learning rate 3×10^{-4}, weight decay 10^{-4}), a cosine learning-rate schedule over 30 epochs, cross-entropy loss, batch size 32, and early stopping on validation loss (patience 7); the checkpoint with the lowest validation loss was retained. Input resolution followed each model's `timm` pretrained configuration, with ImageNet normalization. Training augmentation comprised random horizontal and vertical flips, random rotation of up to 20 degrees, and brightness and

Table 2: Class-by-class binary label mapping and image counts. NILM: negative for intraepithelial lesion or malignancy; ASC-US: atypical squamous cells of undetermined significance; ASC-H: atypical squamous cells, cannot exclude a high-grade lesion; LSIL and HSIL: low-grade and high-grade squamous intraepithelial lesion; SCC: squamous cell carcinoma. Metaplastic cells in SIPaKMeD are benign and mapped to normal; the equivocal ASC-US and ASC-H categories are grouped with abnormal, a convention that is itself debatable.

Dataset	Binary label	Original classes (images)
SIPaKMeD (4,049)	Normal (2,411)	Superficial-intermediate (831), parabasal (787), metaplastic (793)
	Abnormal (1,638)	Koilocytotic (825), dyskeratotic (813)
Herlev (917)	Normal (242)	Normal columnar (98), normal intermediate (70), normal superficial (74)
	Abnormal (675)	Light dysplastic (182), moderate dysplastic (146), severe dysplastic (197), carcinoma in situ (150)
Mendeley LBC (962)	Normal (612)	NILM (612)
	Abnormal (350)	LSIL (113), HSIL (163), SCC (74)
APCEM (4,253)	Normal (779)	NILM (779)
	Abnormal (3,474)	ASC-US (340), ASC-H (119), LSIL (1,241), HSIL (1,300), SSC (474)

contrast jitter of 0.15. Training used a single graphics processing unit (GPU; Table 3). We report balanced accuracy, macro F1, the area under the receiver operating characteristic curve (AUROC), and parameter count. For the five-class in-distribution evaluation, balanced accuracy has a chance level of 0.20, whereas for the binary external evaluation it has a chance level of 0.50, so the two are not on the same scale and are compared only by rank. Evaluation stages were run as separate training and evaluation runs (see Sec. 3.10).

3.3.1 Data split and source-image overlap

SIPaKMeD was split 60/15/10/15 into training, validation, calibration, and test sets. The split is stratified by class (shuffled within each class with seed 13 plus the class index) and is performed at the level of single-cell crops. The 4,049 cells come from 966 cluster images (about 4.2 cells per image), and cells from one cluster image were not kept together. Reproducing the split procedure on the crop filenames shows that 533 of the 607 test cells (87.8%) share a source cluster image with at least one training cell. Patient and slide identifiers are not available, so even cluster-level separation would not guarantee patient-level independence. In-distribution results, including in-distribution calibration, abstention, and degradation results, may therefore be optimistic and should not be read as an estimate of performance on new patients; Sec. 4.4 reports a cluster-grouped re-run that gauges the size of this effect. The external-dataset evaluations are not affected by this overlap, because no external image was used for training, validation, or calibration. The same split manifest was used at all training seeds.

3.4 Cross-Dataset Domain-Shift Evaluation

Each model was evaluated zero-shot on Herlev, Mendeley LBC, and APCEM, with no retraining or recalibration, by summing its five-class softmax output into normal and abnormal probabilities according to Table 2. To measure how much individual rankings depend on the training run, all 12 models were additionally retrained at two further seeds (17 and 29) and evaluated zero-shot on all three external datasets from a single checkpoint per model per seed.

3.5 Explainability Backbone and Methods

Relevancy propagation and attention rollout are defined for standard multi-head self-attention blocks, and EfficientViT-M0 has no established propagation rule for either. Rather than force an ad hoc extension, the explainability and abstention experiments use **DeiT-Tiny**, a standard `timm` `VisionTransformer` that reached 0.9606 balanced accuracy in the benchmark (Sec. 4.1), retrained for these experiments. All three methods produce a relevance map over the 14×14 patch grid (224-pixel input, 16-pixel patches), upsampled to image resolution. *Attention rollout*: raw

softmax attention averaged over heads and chained across all 12 blocks; class-agnostic. *Relevancy propagation*: the same rollout recurrence, with each block's contribution equal to $\mathrm{ReLU}(\nabla_{\mathrm{attn}} \odot \mathrm{attn})$ averaged over heads, where the gradient is taken with respect to the model's predicted-class logit; class-discriminative. *Grad-CAM (ViT-adapted)*: gradient-weighted channel importance applied to a transformer block's token activations. The target layer must be the *second-to-last* block: DeiT-Tiny pools only the classification (CLS) token, so no layer after the final block mixes information back into patch positions. We verified this directly: every patch token's gradient at the last block's output was exactly zero, whereas the second-to-last block's patch-token gradients vary normally. Targeting the wrong block therefore yields an all-zero heatmap silently, not an error.

3.6 Faithfulness and Localization Metrics

Deletion/insertion area under the curve (AUC)[29]: patches are ranked by a method's own relevance score and progressively replaced with (deletion) or revealed from (insertion) a heavily Gaussian-blurred copy of the image, tracking the predicted-class probability; we report the area under the probability-versus-fraction curve (lower is better for deletion, higher for insertion). *Nucleus concentration ratio*: SIPaKMeD provides expert-marked nucleus boundary polygons per cell, which we persist in crop-local pixel coordinates. The ratio is the fraction of a heatmap's total relevance mass inside the nucleus divided by the nucleus's share of image area. A ratio of 1.0 is exactly chance, and higher values mean relevance concentrates in the nucleus more than chance would predict, which is clinically the correct place to concentrate.

3.7 Calibration, Conformal Prediction, and the Combined Abstention Signal

Temperature scaling[33] was fit on a calibration split disjoint from training and test data. Conformal prediction sets used the nonconformity score one minus the softmax probability of the true class, a target miscoverage of 0.1, and the calibration split (about 400 images, 10% of SIPaKMeD, which shares source images with the training split as described in Sec. 3.3.1). The threshold was the mean, over 200 bootstrap resamples of the calibration scores (each 80% of the calibration set, drawn with replacement), of the finite-sample-corrected split-conformal quantile at level $\lceil (n+1)(1-\alpha) \rceil / n$, and a set contains every class whose softmax probability is at least one minus the threshold. This averaged threshold is a stabilizing heuristic for the small calibration set and does not carry the finite-sample guarantee of a single split-conformal threshold, and the standard single split-conformal threshold was not evaluated for comparison.

3.8 Efficiency and Latency Measurement

Parameters and FLOPs (via `torch.utils.flop_counter.FlopCounterMode`, counting a multiply-add as two FLOPs) are architecture properties independent of trained weights, so they were measured without training. CPU latency (single-image batch of random input, PyTorch eager mode, median of 30 iterations after 5 warm-up iterations) and memory (isolated per model in its own subprocess) were likewise measured with untrained weights, on one shared consumer laptop. Untrained weights are appropriate because the latency of dense inference depends on the architecture and tensor shapes rather than on learned weight values. The laptop's CPU model, thread count, and library versions were not recorded (Table 3).

3.9 Use of Generative AI Tools

Generative AI tools assisted with manuscript drafting and language refinement, under the author's direction, including merging and restructuring text from the author's two earlier manuscripts and cross-checking reported numbers against the source tables.

3.10 Note on Run-to-Run Variation

In-distribution accuracy of the same model differs slightly between tables (for example, EfficientViT-M0: 0.9654 in Table 4, 0.9737 in Table 5, 0.9621 in Table 6, and about 0.975 in the degradation experiment; all are five-class balanced accuracy). The benchmark, the Herlev and Mendeley LBC evaluation, and the APCEM evaluation each used the same recipe and seed (13), but each model was trained afresh in a separate session for each stage; ViT-Small's Herlev score, for example, was 0.7194 in a partially lost earlier run and 0.7049 in the retrained run. The training script sets only the PyTorch random seed; it does not enable deterministic cuDNN algorithms and does not separately seed NumPy, Python, or the four shuffled data-loader workers, so identically seeded runs are not expected to be bit-identical. We did not isolate which factor accounts for the differences, so we treat them as run-to-run variation of unidentified origin. The three-seed analysis (Sec. 4.3) measures sensitivity to the training seed, whereas same-seed repeatability was not measured separately. The degradation experiment additionally used a different random train/validation split. Comparisons are therefore valid within a table, not across tables.

3.11 Statistical Analysis

Per-image predictions were not retained for the original runs, so bootstrap confidence intervals could not be computed for them. The cluster-grouped re-run (Sec. 4.4) retained them, and for that run we report 95% bootstrap confidence intervals (2,000 resamples) for balanced accuracy, resampling source cluster images for in-distribution results and individual images for external results,

Table 3: Reproducibility information. Items marked "not recorded" could not be recovered from the repository.

Component	Information
Software versions	Cluster-grouped re-run: Python 3.13.15, torch 2.11.0+cu128, timm 1.0.29, numpy 2.1.3, CUDA 12.8, cuDNN 9.19. Original runs: minimum versions only (torch $\geq$2.2, timm $\geq$1.0); exact versions not recorded
Training hardware	Google Colab T4 GPU (original benchmark; other stages not individually recorded); Kaggle Tesla T4 for the cluster-grouped re-run (12 models in 1.6 GPU hours)
Latency hardware	One shared consumer laptop (CPU model and thread count not recorded); explanation-cost pair on a cloud virtual-machine CPU
Random seeds	Training seeds 13 (all stages), 17 and 29 (seed check); split seed 13 plus class index
Determinism	Only the PyTorch seed is set; deterministic cuDNN not enabled; NumPy, Python, and data-loader workers not separately seeded
Data split	One stratified cell-level manifest, identical across training seeds (Sec. 3.3.1)
Checkpoint selection	Lowest validation loss, early stopping (patience 7, at most 30 epochs)

together with per-class sensitivity and specificity. For the original runs we instead report three things. First, the mean and standard deviation of balanced accuracy over three training seeds (13, 17, and 29) for every model and dataset (Table 7). Second, Spearman rank correlations across the 12 models between three-seed mean external balanced accuracy and in-distribution accuracy, parameter count, FLOPs, and architecture family (CNN versus transformer; Table 8). With 12 models, $|\rho| \geq 0.587$ is required for two-sided $p < 0.05$ without correction for the twelve comparisons, so we report these correlations descriptively. Third, conservative worst-case 95% half-widths for test-set sampling error of a single balanced accuracy, $1.96 \times 0.5\sqrt{0.25/n_{\text{normal}} + 0.25/n_{\text{abnormal}}}$, which assume independent images and the largest possible variance. Image-level sampling uncertainty and training-seed variability are different sources of uncertainty and are reported separately.

4 Results

4.1 Efficiency–Accuracy Benchmark

The two smallest models in the registry were also the two most accurate (Table 4, Fig. 1). EfficientViT-M0 exceeded the three large baselines, Swin-Tiny (27.5M parameters), ResNet-50 (23.5M), and ViT-Small (21.7M), by 0.5, 1.1, and 2.8 points of balanced accuracy respectively. On this task, at this data scale, the efficiency–accuracy trade-off does not appear in distribution. This is a single-seed result on a cell-level split with substantial source-image overlap (Sec. 3.3.1), so differences of a few points among models near 0.96 should not be over-interpreted. Across three seeds (Table 7), EfficientViT-M0 and EfficientViT-B0 ranked second and third by mean in-distribution accuracy

Table 4: In-distribution benchmark on SIPaKMeD, all 12 models, sorted by balanced accuracy (seed 13).

Model	Bal. acc.	Macro F1	AUROC	Params
EfficientViT-M0	**0.9654**	**0.9656**	0.9985	**2,157,401**
EfficientViT-B0	0.9623	0.9624	0.9983	2,135,413
DeiT-Tiny	0.9606	0.9607	0.9974	5,525,381
Swin-Tiny	0.9605	0.9606	0.9965	27,523,199
TinyViT-5M	0.9587	0.9589	0.9979	5,073,369
MobileViT-S	0.9570	0.9575	0.9979	4,940,837
ResNet-50	0.9542	0.9540	0.9966	23,518,277
LeViT-128	0.9458	0.9456	0.9946	8,447,786
MobileViTv2	0.9387	0.9401	0.9966	4,391,406
EfficientNet-Lite0	0.9375	0.9376	0.9944	3,377,413
ViT-Small	0.9374	0.9380	0.9960	21,667,589
MobileNetV2	0.9357	0.9365	0.9952	2,230,277

(0.966 each, behind TinyViT-5M at 0.968), with seed standard deviations of 0.008–0.009 that are as large as the gaps among the leaders, so the specific ordering at the top of Table 4 is not stable across seeds.

4.2 Cross-Dataset Domain Shift

4.2.1 In-distribution accuracy did not predict robustness consistently

The two most accurate models in distribution in this run (EfficientViT-B0, 0.9738, and EfficientViT-M0, 0.9737) ranked 9th and 10th of 12 on Herlev and 6th and 10th of 12 on Mendeley LBC (Table 5). The comparison of EfficientViT-M0 with DeiT-Tiny is the sharpest case: the better in-distribution model (0.9737 versus 0.9504) was worse under both shifts (Herlev 0.6716 versus 0.7151; Mendeley LBC 0.4731 versus 0.5230, below the 0.50 chance floor). The same reversal appeared on APCEM (Table 6), where EfficientViT-M0 again scored lower under shift than DeiT-Tiny (0.3730 versus 0.4469) despite higher in-distribution accuracy (0.9621 versus 0.9589).

4.2.2 A family-level pattern appears on Mendeley LBC

All three CNNs scored 0.539–0.667, and eight of the nine transformer-family models scored 0.293–0.523. The ninth, MobileViTv2, scored 0.5418, above the lowest CNN (0.5391) by 0.003, a gap the seed check shows is noise (Sec. 4.3). The split does not hold on Herlev, where transformer models occupy both the best (MobileViTv2, TinyViT-5M) and the weakest (LeViT-128) positions.

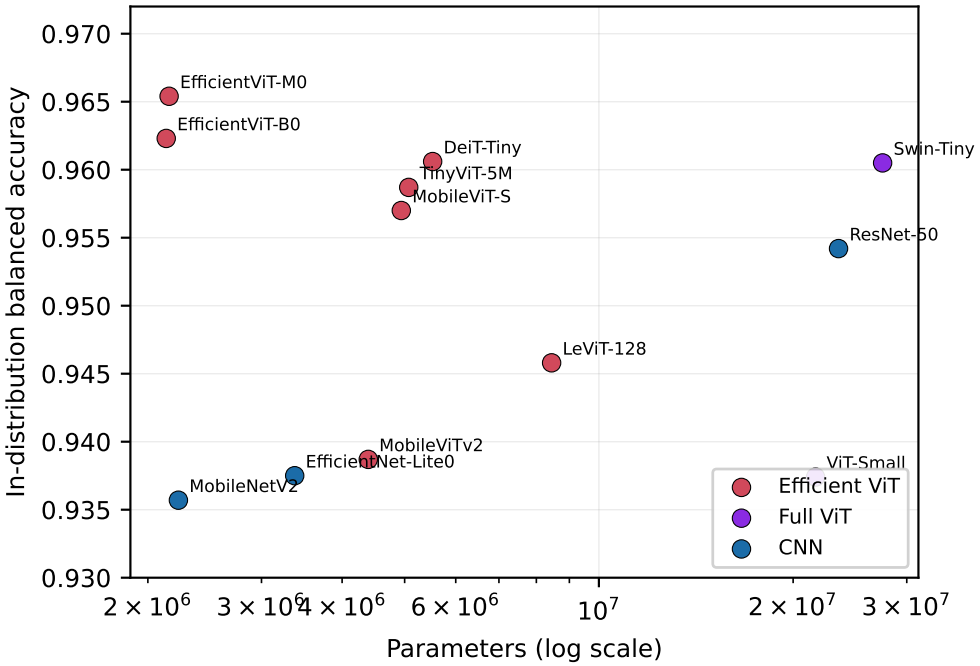


Figure 1: Parameters (log scale) versus in-distribution balanced accuracy, colored by architecture family. The two efficient-ViT models with the fewest parameters (EfficientViT-M0 and -B0) are also the two most accurate in this run.

Table 5: Balanced accuracy of all 12 models, seed 13, sorted by Mendeley LBC score. The SIPaKMeD column is five-class in-distribution balanced accuracy from the same run (chance 0.20); the Herlev and Mendeley LBC columns are zero-shot binary normal/abnormal balanced accuracy (chance 0.50). The scales are not comparable, so only ranks are compared across columns.

Model (family, params)	SIPaKMeD (5-class)	Herlev	Mendeley LBC
EfficientNet-Lite0 (CNN, 3.38M)	0.9455	0.5970	**0.6674**
ResNet-50 (CNN, 23.5M)	0.9556	0.6838	0.6082
MobileViTv2 (eff. ViT, 4.39M)	0.9638	**0.7501**	0.5418
MobileNetV2 (CNN, 2.23M)	0.9505	0.6919	0.5391
DeiT-Tiny (eff. ViT, 5.5M)	0.9504	0.7151	0.5230
EfficientViT-B0 (eff. ViT, 2.14M)	0.9738	0.6818	0.5217
Swin-Tiny (full ViT, 27.5M)	0.9590	0.7282	0.5192
ViT-Small (full ViT, 21.7M)	0.9573	0.7049	0.5084
LeViT-128 (eff. ViT, 8.4M)	0.9526	0.5777	0.4764
EfficientViT-M0 (eff. ViT, 2.16M)	0.9737	0.6716	0.4731
TinyViT-5M (eff. ViT, 5.1M)	0.9685	0.7327	0.4543
MobileViT-S (eff. ViT, 4.94M)	0.9520	0.7104	**0.2927** (AUROC 0.215)

4.2.3 *Architecture lineage did not predict robustness*

MobileViT-S scored far below chance on Mendeley LBC (balanced accuracy 0.2927, AUROC 0.215), meaning its predictions were anti-correlated with the ground truth, not merely uninformative. Because Mendeley LBC images are whole fields rather than single cells, this may reflect the scale and content mismatch as much as institutional differences, and we cannot separate these. Its direct successor MobileViTv2 shows no such failure and was the strongest model in the registry on Herlev (0.7501).

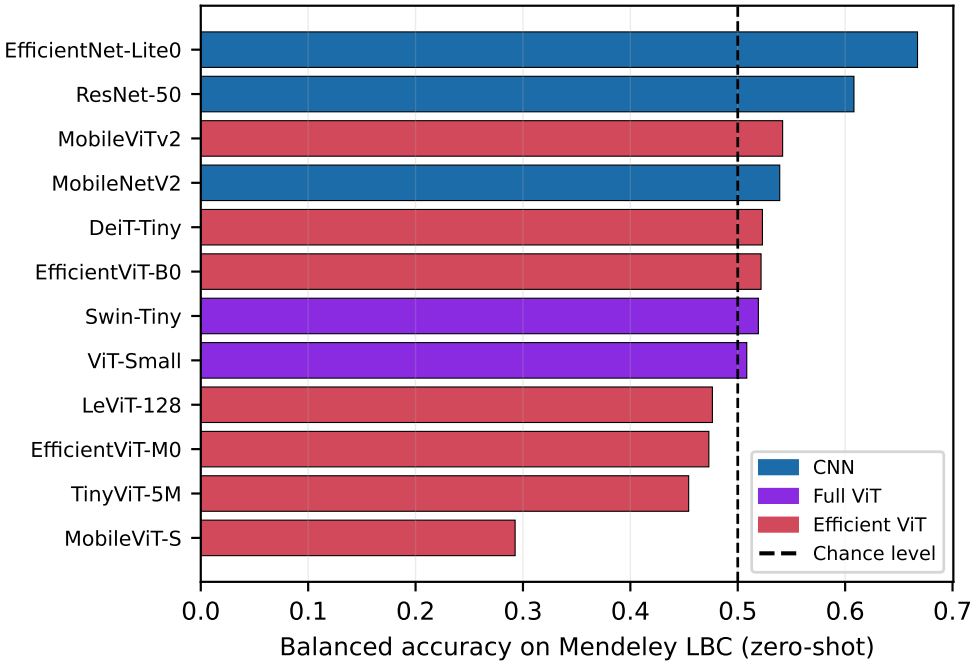


Figure 2: Balanced accuracy on Mendeley LBC (zero-shot), sorted and colored by architecture family. All three CNNs cluster above chance, and eight of nine transformer-family models cluster at or below chance, with MobileViT-S the sole catastrophic outlier.

4.2.4 *A third dataset overturns the family-level pattern*

On APCEM, 9 of 12 models fell below 0.50 on both balanced accuracy and AUROC (Table 6). Only MobileNetV2 (0.6158 balanced accuracy, AUROC 0.6528), EfficientNet-Lite0 (0.5393, 0.5701), and LeViT-128 (0.5263, 0.5376) scored above chance, and MobileNetV2 was the clear best. The CNN family spans both the best model (MobileNetV2) and the model with the lowest balanced accuracy (ResNet-50, 0.3339; AUROC 0.2929, second-lowest of the 12). The CNN-family robustness seen on Mendeley LBC therefore does not survive a third, independently sourced dataset.

Table 6: Zero-shot binary evaluation of all 12 models on APCEM, sorted by AUROC. This table comes from a separate training run, so the five-class in-distribution balanced accuracy in the SIPaKMeD column differs from Table 5.

Model (family, params)	SIPaKMeD (5-class)	APCEM bal. acc.	APCEM AUROC
MobileNetV2 (CNN, 2.23M)	0.9623	**0.6158**	**0.6528**
EfficientNet-Lite0 (CNN, 3.38M)	0.9309	0.5393	0.5701
LeViT-128 (eff. ViT, 8.4M)	0.9001	0.5263	0.5376
Swin-Tiny (full ViT, 27.5M)	0.9330	0.4724	0.3082
EfficientViT-B0 (eff. ViT, 2.14M)	0.9638	0.4481	0.3206
DeiT-Tiny (eff. ViT, 5.5M)	0.9589	0.4469	0.3799
MobileViTv2 (eff. ViT, 4.39M)	0.9570	0.4290	0.4120
TinyViT-5M (eff. ViT, 5.1M)	0.9670	0.4100	0.3681
EfficientViT-M0 (eff. ViT, 2.16M)	0.9621	0.3730	0.3407
MobileViT-S (eff. ViT, 4.94M)	0.9687	0.3621	0.3161
ViT-Small (full ViT, 21.7M)	0.9538	0.3444	0.2555
ResNet-50 (CNN, 23.5M)	0.9639	**0.3339**	0.2929

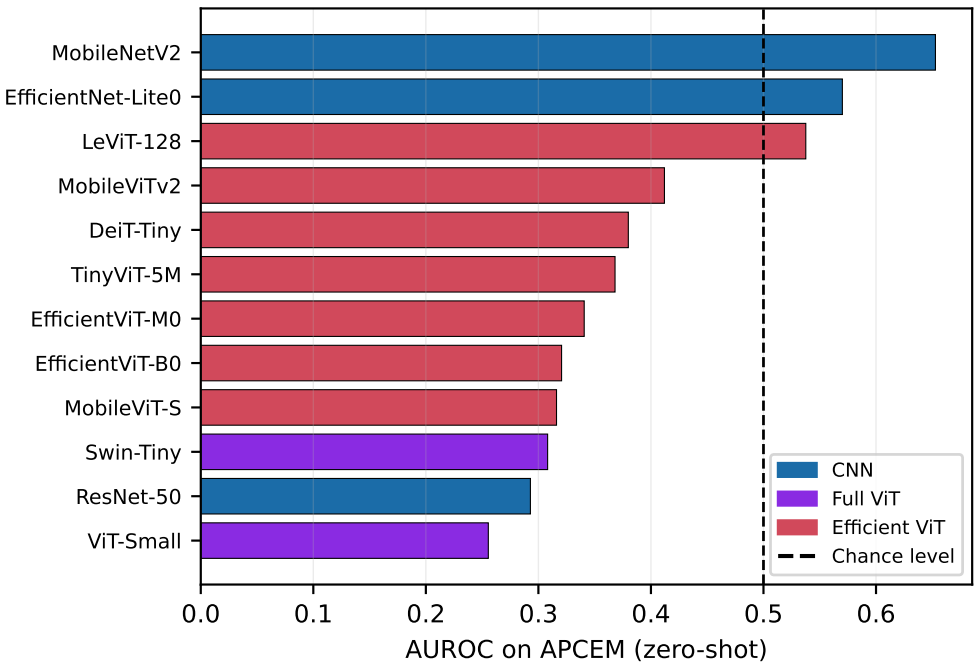


Figure 3: AUROC on APCEM (zero-shot), sorted and colored by architecture family. Unlike Mendeley LBC, the CNN family spans both the best (MobileNetV2) and one of the worst (ResNet-50) models; no family-level pattern survives this third dataset.

4.3 Seed Sensitivity

Variance across random seeds is a known threat to benchmark conclusions.[56,57] Each headline model was trained once at seed 13. We measured seed-to-seed variance directly for two close-ranked pairs at two further seeds (17 and 29), and then extended the check to all 12 models on all three external datasets.

4.3.1 Close calls can go either way

On Mendeley LBC, MobileViTv2's 0.003 edge over MobileNetV2 (0.5418 versus 0.5391) does not survive: MobileNetV2 ranges 0.493–0.604 and MobileViTv2 ranges 0.494–0.623 across the three seeds, the ranges overlap almost completely, and the ranking flips at seed 17. On APCEM, EfficientViT-B0's 0.001 edge over DeiT-Tiny (0.4481 versus 0.4469) does survive: EfficientViT-B0 ranges 0.421–0.448, DeiT-Tiny ranges 0.367–0.447, EfficientViT-B0 wins at all three seeds, and its margin widens (to 0.027, then 0.067). The same kind of close call produced opposite outcomes once tested.

4.3.2 The Mendeley LBC family split is seed-specific

At seed 13 only 1 of 9 transformer-family models exceeds the weakest CNN (by 0.003). At seed 17, 3 of 9 do, and at seed 29, 4 of 9 do, by margins up to 0.11.

4.3.3 The in-distribution leader is not reliably the robustness loser

EfficientViT-M0 is the (near-)in-distribution leader and ranks 10th of 12 on both Herlev and Mendeley LBC at seed 13. At seed 29 it is the outright in-distribution winner and mid-pack under shift on both datasets (rank 4–5 of 12), and at seed 17 it is not even the in-distribution winner (rank 10 of 12). By three-seed mean, EfficientViT-M0 ranked second in distribution but 8th, 7th, and 8th of 12 on Herlev, Mendeley LBC, and APCEM. It should be read as a seed-13 illustration rather than a reproducible property of that model.

4.3.4 Some findings replicate cleanly

Herlev's inverse pattern (most transformer models outscoring the CNNs) and MobileNetV2's status as the strongest model on APCEM both replicate across all three seeds.

4.3.5 Rank correlations with static properties

Across the 12 models, the association of three-seed mean external balanced accuracy with the static properties we measured was not consistent across datasets (Table 8). FLOPs correlated

Table 7: Mean ± standard deviation of balanced accuracy over three training seeds (13, 17, 29), in the row order of Table 5. The SIPaKMeD column is five-class; the external columns are binary. Seed 13 values are those of Tables 5 and 6, which come from separate runs.

Model	SIPaKMeD (5-class)	Herlev	Mendeley LBC	APCEM
EfficientNet-Lite0	0.951 ± 0.006	0.630 ± 0.030	0.630 ± 0.106	0.544 ± 0.027
ResNet-50	0.957 ± 0.002	0.689 ± 0.005	0.537 ± 0.062	0.390 ± 0.072
MobileViTv2	0.964 ± 0.004	0.721 ± 0.025	0.516 ± 0.064	0.466 ± 0.033
MobileNetV2	0.946 ± 0.004	0.659 ± 0.030	0.563 ± 0.083	0.586 ± 0.026
DeiT-Tiny	0.955 ± 0.004	0.700 ± 0.015	0.484 ± 0.036	0.385 ± 0.059
EfficientViT-B0	0.966 ± 0.008	0.667 ± 0.017	0.575 ± 0.047	0.442 ± 0.035
Swin-Tiny	0.961 ± 0.005	0.720 ± 0.009	0.481 ± 0.034	0.419 ± 0.047
ViT-Small	0.957 ± 0.008	0.705 ± 0.048	0.477 ± 0.055	0.392 ± 0.045
LeViT-128	0.900 ± 0.065	0.538 ± 0.034	0.507 ± 0.031	0.518 ± 0.016
EfficientViT-M0	0.966 ± 0.009	0.681 ± 0.015	0.502 ± 0.038	0.409 ± 0.043
TinyViT-5M	0.968 ± 0.001	0.706 ± 0.025	0.453 ± 0.010	0.422 ± 0.018
MobileViT-S	0.958 ± 0.005	0.689 ± 0.029	0.387 ± 0.084	0.372 ± 0.024

positively with Herlev accuracy ($+0.68$) but negatively with Mendeley LBC (-0.55) and APCEM (-0.48) accuracy. In-distribution accuracy had a moderate positive association on Herlev ($+0.55$) and weak negative associations on the other two datasets. Architecture family mattered most on Mendeley LBC (CNN versus transformer, $+0.64$) and much less on APCEM ($+0.36$) and Herlev (-0.42). The two correlations that pass the uncorrected threshold (FLOPs on Herlev and family on Mendeley LBC) would not survive correction for the twelve comparisons, and each weakens or reverses on another dataset. We therefore conclude only that these properties did not predict robustness consistently across these three datasets, not that they are unrelated to robustness on any single dataset.

Table 8: Spearman rank correlation ($n = 12$ models) between three-seed mean external balanced accuracy and static properties. Family is coded CNN $= 1$, transformer $= 0$. $|\rho| \geq 0.587$ corresponds to two-sided $p < 0.05$ without correction for multiple comparisons.

Property	Herlev	Mendeley LBC	APCEM
In-distribution accuracy (five-class)	$+0.55$	-0.31	-0.27
Parameters	$+0.43$	-0.46	-0.41
FLOPs	$+0.68$	-0.55	-0.48
Family (CNN versus transformer)	-0.42	$+0.64$	$+0.36$

4.3.6 Sampling uncertainty

Conservative worst-case 95% half-widths for the balanced accuracy of a single model are ± 0.037 on Herlev (242 normal, 675 abnormal images), ± 0.033 on Mendeley LBC (612, 350), and ± 0.019 on APCEM (779, 3,474), before accounting for training-seed variance. Differences smaller than

these, such as the 0.003 gap between MobileViTv2 and MobileNetV2 on Mendeley LBC, are not distinguishable from sampling error, and many neighboring ranks in Tables 5 and 6 are not distinguishable either. The seed standard deviations in Table 7 (median 0.025 on Herlev, 0.051 on Mendeley LBC, and 0.034 on APCEM) are of similar or larger size.

4.4 Cluster-Grouped Re-Run

To gauge the effect of the source-image overlap, all 12 models were retrained once (seed 13, the same recipe) on a cluster-grouped split in which every source cluster image lies in exactly one split (2,439 training, 630 validation, 384 calibration, and 596 test cells; 163 test clusters) and evaluated zero-shot on the three external datasets, with per-image predictions retained (Table 9). This is a single run, and the comparison with the earlier cell-level run is unpaired (different test cells), so it estimates the size of the effect only roughly.

Table 9: Cluster-grouped re-run (seed 13). The SIPaKMeD column is five-class balanced accuracy with a 95% cluster-bootstrap interval; Δ is the difference from the cell-level seed-13 value in Table 5 (unpaired). External columns are binary balanced accuracy with 95% image-bootstrap intervals. Rows are in the order of Table 5.

Model	SIPaKMeD (5-class)	Δ	Herlev	Mendeley LBC	APCEM
EfficientNet-Lite0	0.953 [0.93, 0.97]	+0.008	0.540 [0.51, 0.57]	0.632 [0.60, 0.66]	0.520 [0.50, 0.54]
ResNet-50	0.961 [0.94, 0.98]	+0.006	0.721 [0.69, 0.76]	0.541 [0.51, 0.57]	0.516 [0.50, 0.54]
MobileViTv2	0.968 [0.95, 0.98]	+0.004	0.722 [0.69, 0.75]	0.487 [0.46, 0.52]	0.410 [0.39, 0.43]
MobileNetV2	0.912 [0.88, 0.94]	−0.038	0.633 [0.60, 0.67]	0.531 [0.50, 0.56]	0.648 [0.63, 0.67]
DeiT-Tiny	0.958 [0.94, 0.98]	+0.008	0.730 [0.70, 0.76]	0.460 [0.44, 0.48]	0.434 [0.41, 0.45]
EfficientViT-B0	0.953 [0.93, 0.97]	−0.021	0.625 [0.59, 0.66]	0.559 [0.53, 0.59]	0.426 [0.41, 0.44]
Swin-Tiny	0.951 [0.93, 0.97]	−0.008	0.747 [0.71, 0.78]	0.436 [0.42, 0.46]	0.332 [0.31, 0.35]
ViT-Small	0.935 [0.91, 0.96]	−0.022	0.715 [0.68, 0.75]	0.443 [0.42, 0.47]	0.498 [0.49, 0.51]
LeViT-128	0.807 [0.75, 0.86]	−0.146	0.538 [0.51, 0.57]	0.504 [0.50, 0.51]	0.510 [0.51, 0.51]
EfficientViT-M0	0.947 [0.92, 0.97]	−0.027	0.655 [0.62, 0.69]	0.448 [0.42, 0.47]	0.441 [0.42, 0.46]
TinyViT-5M	0.958 [0.94, 0.98]	−0.010	0.703 [0.67, 0.74]	0.418 [0.39, 0.45]	0.434 [0.43, 0.44]
MobileViT-S	0.963 [0.95, 0.98]	+0.011	0.659 [0.63, 0.69]	0.416 [0.39, 0.44]	0.476 [0.47, 0.49]

In distribution, the median change from the cell-level split across the 12 models was -0.9 points of balanced accuracy, with changes from -14.6 to $+1.1$ points. The largest change, for LeViT-128, reflects a training failure on this split (early stopping at epoch 8, balanced accuracy 0.807) in a model that was already unstable across seeds (Table 7: 0.900 ± 0.065), not a leakage effect; excluding it the median is -0.8 points. These changes are of the size of the training-seed standard deviations (median 0.5 points, Table 7) and of the cluster-bootstrap interval half-widths (about ± 2 points), and cluster-level resampling widened the intervals by about 27% relative to image-level resampling (mean width 0.042 versus 0.033). A single re-run therefore does not show a large inflation of in-distribution accuracy from the overlap, and it does not exclude a small one.

On the external datasets the re-run followed the same pattern as the original runs: Herlev balanced accuracy was highest (0.54–0.75), and Mendeley LBC and APCEM were mostly at or be-

low 0.5. Per-class results show that much of the below-chance performance on the field-level datasets reflects near one-class prediction. For example, on APCEM ViT-Small had sensitivity 0.07 and specificity 0.93, whereas TinyViT-5M had sensitivity 0.84 and specificity 0.03. Per-class sensitivity, specificity, and confusion matrices for every model and dataset are in the repository (`runs/kaggle_cluster_split/results.json`).

4.5 Explainability Fidelity

Table 10: Explainability method comparison on DeiT-Tiny, 50 SIPaKMeD test images (10 per class), all matched to a real nucleus polygon.

Method	Deletion AUC↓	Insertion AUC↑	Nucleus conc.↑
Relevancy propagation	**0.734**	**0.970**	**2.714**
Grad-CAM (ViT-adapted)	0.818	0.877	1.472
Attention rollout	0.825	0.936	1.177

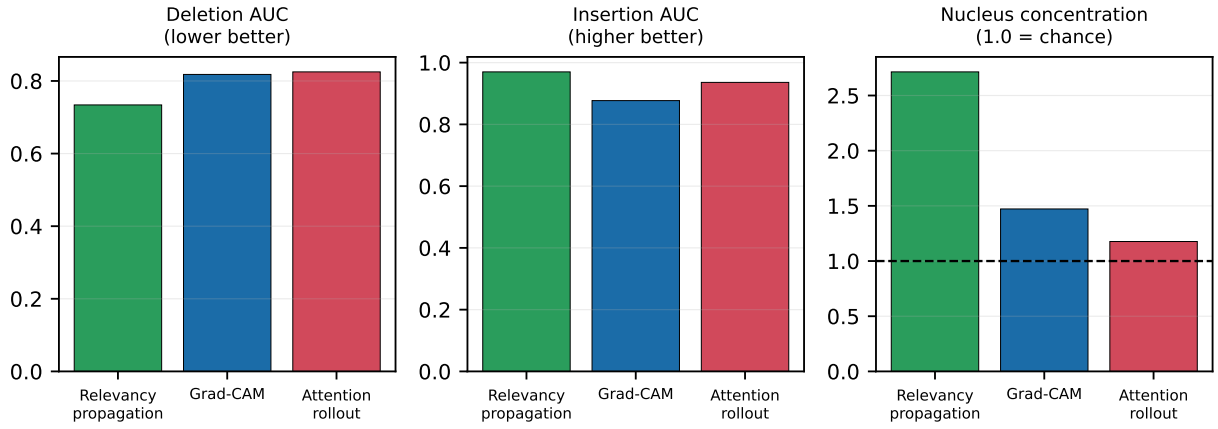


Figure 4: Deletion AUC, insertion AUC, and nucleus concentration ratio for the three explainability methods. Relevancy propagation wins on all three, with the widest margin on nucleus concentration.

Relevancy propagation won every comparison (Table 10), by a wide margin on nucleus concentration (2.71 times chance versus attention rollout's 1.18 times, barely above chance). These are means over 50 cells from the original cell-level run, for which per-cell scores were not retained; Sec. 4.5.2 repeats the comparison on all cluster-grouped test cells with paired intervals. We attribute the gap to a plausible mechanistic difference rather than noise: attention rollout is class-agnostic, a disadvantage on an inherently class-discriminative test, whereas relevancy propagation is class-discriminative and integrates the gradient-weighted signal across all twelve blocks.

4.5.1 *Second backbone*

This single-backbone result rested on DeiT-Tiny, so we repeated the comparison on ViT-Small, a structurally identical `VisionTransformer` (same `global_pool="token"`, `num_prefix_tokens=1`, 12 blocks, verified directly), trained fresh with the same recipe and seed.

Table 11: Explainability ranking on a second backbone (ViT-Small) versus the original (DeiT-Tiny).

	DeiT-Tiny (original)			ViT-Small (2nd backbone)		
Method	Del.↓	Ins.↑	Conc.↑	Del.↓	Ins.↑	Conc.↑
Relevancy prop.	**0.734**	**0.970**	**2.71**	**0.816**	**0.965**	**4.20**
Grad-CAM	0.818	0.877	1.47	0.888	0.895	1.35
Attn. rollout	0.825	0.936	1.18	0.888	0.927	0.97

The ranking replicated exactly (Table 11): relevancy propagation was the most faithful of the three methods on all three metrics on both evaluated backbones, with an even wider nucleus-concentration margin (4.20 times chance versus its closest competitor's 1.35 times). The claim is scoped deliberately. Both backbones are pure multi-head self-attention transformers of different widths and depths, not different architecture families, and whether the ranking survives a hybrid convolution–attention backbone remains untested because relevancy propagation is not defined for EfficientViT-M0's architecture.

4.5.2 *Full-sample re-evaluation on the cluster-grouped checkpoints*

We repeated the comparison with the cluster-grouped DeiT-Tiny and ViT-Small checkpoints (test balanced accuracy 0.958 and 0.935) on all 596 cluster-grouped test cells, every one of which has an expert nucleus polygon, retaining the per-cell scores. Intervals are 95% cluster-bootstrap intervals (2,000 resamples of the 163 source cluster images). Table 12 gives the means, and Table 13 gives the paired per-cell differences between relevancy propagation and each baseline.

Table 12: Explainability scores on all 596 cluster-grouped test cells, mean [95% cluster-bootstrap interval]. Lower is better for deletion AUC, and higher is better for insertion AUC and nucleus concentration (1.0 is chance).

Backbone	Method	Deletion AUC	Insertion AUC	Nucleus concentration
DeiT-Tiny	Relevancy propagation	0.727 [0.678, 0.777]	0.966 [0.957, 0.973]	3.79 [2.80, 4.81]
DeiT-Tiny	Attention rollout	0.807 [0.768, 0.845]	0.941 [0.925, 0.955]	1.35 [1.15, 1.59]
DeiT-Tiny	Grad-CAM	0.816 [0.779, 0.852]	0.904 [0.876, 0.931]	1.19 [1.01, 1.37]
ViT-Small	Relevancy propagation	0.721 [0.653, 0.789]	0.961 [0.952, 0.970]	4.79 [3.49, 6.17]
ViT-Small	Attention rollout	0.766 [0.699, 0.831]	0.947 [0.932, 0.959]	1.95 [1.69, 2.22]
ViT-Small	Grad-CAM	0.910 [0.886, 0.931]	0.780 [0.714, 0.843]	1.17 [0.91, 1.43]

Table 13: Paired per-cell difference, relevancy propagation minus the baseline, mean [95% cluster-bootstrap interval] (percentage of cells on which relevancy propagation is better). Negative deletion differences and positive insertion and concentration differences favor relevancy propagation.

Backbone	Comparison	Deletion AUC	Insertion AUC	Nucleus concentration
DeiT-Tiny	vs. attention rollout	-0.080 [-0.097, -0.063] (88%)	+0.025 [+0.017, +0.033] (74%)	+2.44 [+1.62, +3.29] (78%)
DeiT-Tiny	vs. Grad-CAM	-0.089 [-0.121, -0.059] (76%)	+0.061 [+0.037, +0.086] (68%)	+2.60 [+1.59, +3.62] (76%)
ViT-Small	vs. attention rollout	-0.045 [-0.060, -0.029] (73%)	+0.015 [+0.010, +0.020] (77%)	+2.84 [+1.78, +3.97] (74%)
ViT-Small	vs. Grad-CAM	-0.189 [-0.254, -0.124] (79%)	+0.182 [+0.119, +0.246] (79%)	+3.62 [+2.25, +5.12] (78%)

Relevancy propagation was better than both baselines on all three metrics on both backbones, and all 12 paired intervals excluded zero (Table 13); it was better on 68–88% of individual cells. On DeiT-Tiny its nucleus concentration was 3.79 times chance, compared with 1.35 for attention rollout and 1.19 for Grad-CAM. The ranking is therefore supported by a paired analysis on a much larger sample than the original 50 cells, and it holds on the cluster-grouped split. Representative cells (Fig. 5) show that a high concentration ratio does not always mean attention to the nucleus itself: in some correctly classified cells relevancy propagation concentrates on the nucleus, whereas in others it highlights the surrounding cytoplasm or the region around the nucleus.

4.6 Selective Prediction

Two proposed sub-claims about safe composition hold by construction in our post-hoc-attachment architecture and were verified rather than assumed: temperature scaling changed zero argmax predictions on the calibration split, and explanation computation is a read-only analysis that cannot alter model weights.

Table 14: Selective-prediction risk-coverage AUC (lower is better) on DeiT-Tiny: confidence alone, conformal set size alone, and the combined signal.

Condition	Confidence only	Set size only	Combined
In-distribution	**0.0050**	0.0834	0.0173
Herlev (moderate shift)	**0.1190**	0.3281	0.1793
Mendeley LBC (severe shift)	0.3244	0.4395	**0.2855**

Composition is not a uniform win (Table 14). In distribution and under moderate shift, confidence alone is already the best individual signal, and combining it with the noisier, discrete conformal-set-size signal dilutes it. Under severe shift the combined signal is the best of the three, beating confidence alone (0.2855 versus 0.3244). Combining therefore helps exactly where the better individual signal was already struggling and hurts elsewhere, so an evaluation at one shift severity would report the result backwards. Even the best signal remains far from a safe abstention

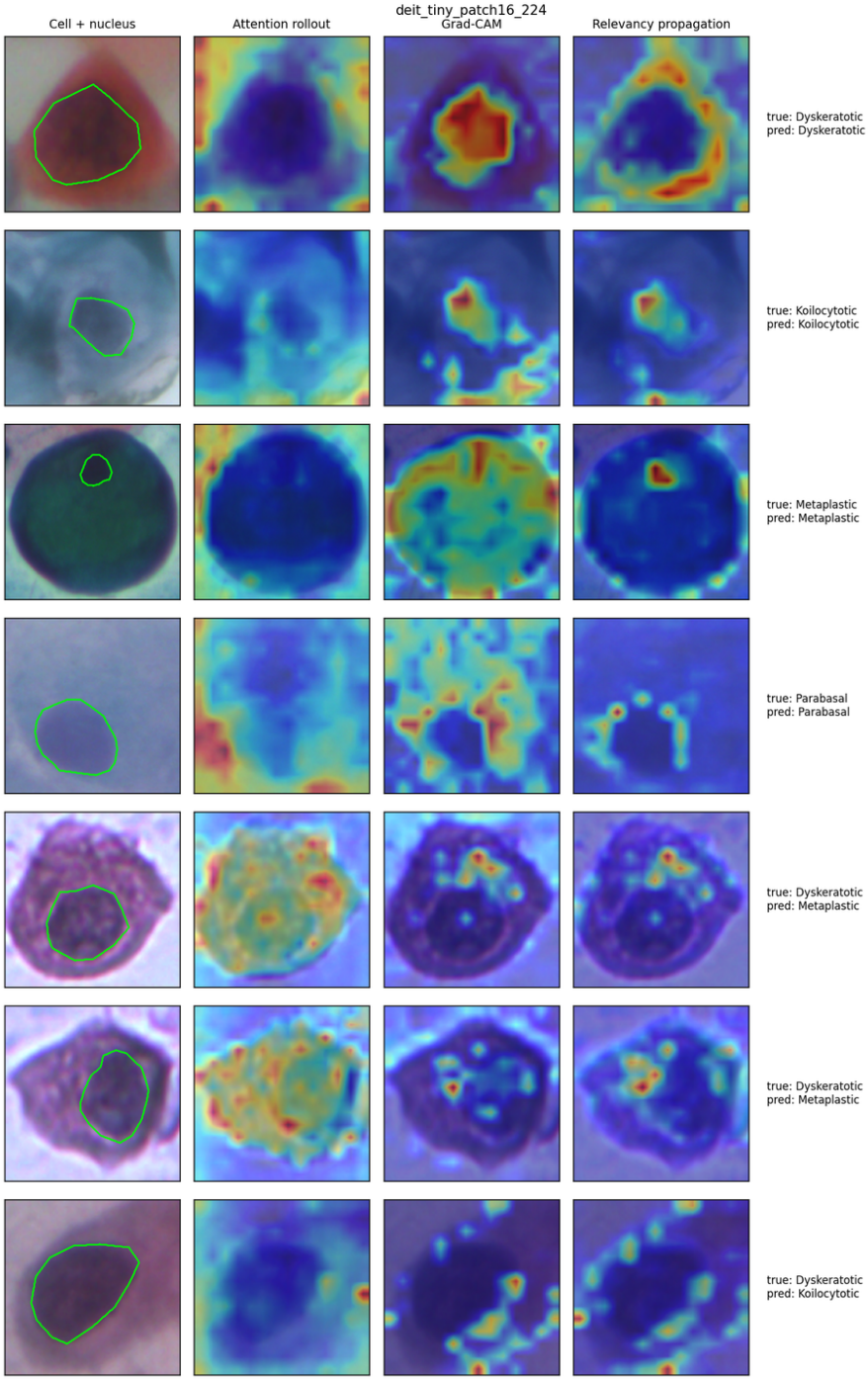


Figure 5: Representative cluster-grouped test cells for DeiT-Tiny: the cell with its expert nucleus outline, and the attention rollout, Grad-CAM, and relevancy propagation heatmaps. The first four rows are the first correctly classified cell of each of the first four classes in test order (the Superficial-Intermediate class is omitted because heatmaps were stored for only the first 400 test cells), and the last three rows are the first misclassified cells; cells were not chosen for appearance. The ViT-Small figure is in the repository (`runs/kaggle_cluster_split_explain/`).

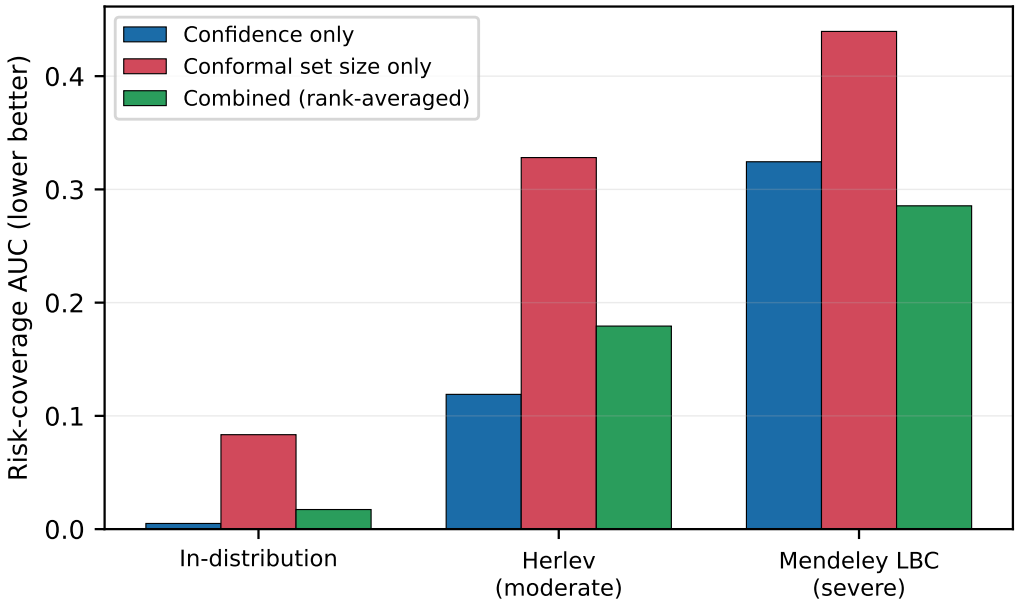


Figure 6: Risk-coverage AUC for confidence alone, conformal set size alone, and the rank-averaged combination at three conditions. Combining hurts in distribution and under moderate shift but helps under severe shift.

rule under severe shift. The in-distribution values are likely optimistic because of the source-image overlap between the calibration, training, and test splits (Sec. 3.3.1), and the conclusion rests on one backbone (DeiT-Tiny).

4.6.1 Operating points on the cluster-grouped re-run

Using the cluster-grouped DeiT-Tiny model and the 384-image calibration split, the standard split-conformal threshold (target miscoverage 0.1, temperature 1.6) gave 89.1% coverage with an average set size of 0.90 on the in-distribution test cells, and the bootstrap-averaged threshold used earlier gave 89.8% coverage with an average set size of 0.91, so the stabilizing heuristic changed the sets little. Table 15 gives the error rate among accepted cases at fixed coverage when abstaining on the least confident cases.

In distribution, abstaining on the least confident 20% of cases reduced the error from 4.2% to 0.4%. Under cross-dataset conditions abstention helped little: on Herlev the error fell from 22.9% to 19.2% at 80% coverage, on Mendeley LBC from 44.4% to 42.2%, and on APCEM the error rose from 64.5% to 66.7%, so confidence was not informative there. These operating points use confidence alone; the combined signal was not recomputed for the re-run.

Table 15: Error rate (%) among accepted cases at fixed coverage for the cluster-grouped DeiT-Tiny model, abstaining on the least confident cases (temperature-scaled maximum probability). In-distribution is five-class; external columns are binary. Lower is better.

Coverage	In-distribution	Herlev	Mendeley LBC	APCEM
100%	4.2	22.9	44.4	64.5
90%	0.6	20.5	43.8	65.8
80%	0.4	19.2	42.2	66.7
70%	0.0	17.9	40.4	68.3

4.7 Resource Efficiency and Deployment Cost

Table 16: Parameters, FLOPs (multiply-add counted as 2), and median single-image CPU latency for all 12 models (untrained weights, one shared consumer laptop).

Model	Params	FLOPs (G)	CPU latency (ms)
EfficientViT-M0	2,157,401	0.158	36.2
EfficientViT-B0	2,135,413	0.198	41.6
MobileNetV2	2,230,277	0.599	33.5
EfficientNet-Lite0	3,377,413	0.768	**137.7**
LeViT-128	8,447,786	0.809	17.5
DeiT-Tiny	5,525,381	2.149	**15.4**
TinyViT-5M	5,073,369	2.510	44.9
MobileViT-S	4,940,837	3.645	66.3
MobileViTv2	4,391,406	3.623	65.2
ResNet-50	23,518,277	8.174	32.0
ViT-Small	21,667,589	8.482	26.8
Swin-Tiny	27,523,199	8.980	54.0

Neither parameter count nor FLOPs reliably predicted CPU latency (Table 16). EfficientNet-Lite0 (3.38M parameters, 0.77 GFLOPs) had more than four times ResNet-50's latency (137.7 ms versus 32.0 ms) despite ResNet-50 having roughly ten times the FLOPs. The two "Mobile"-branded transformers had the highest transformer-family latency (66.3 and 65.2 ms) despite 3.6 GFLOPs, well below Swin-Tiny's 9.0 GFLOPs at 54.0 ms. EfficientViT-M0, the leader on parameters and FLOPs, was not the fastest model in wall-clock terms: DeiT-Tiny (15.4 ms) and LeViT-128 (17.5 ms) were both faster despite having about 14 and 5 times more FLOPs respectively. Repeating the EfficientViT-M0 measurement in isolation three times gave 27.4, 28.4, and 27.9 ms, against 28.3 and 36.2 ms in two full sweeps, so individual latencies carry roughly a 30% run-to-run swing from background load. The relative ordering (EfficientNet-Lite0 far slower than all others, DeiT-Tiny and LeViT-128 fastest) was stable across repeated runs.

4.7.1 *Cost of an explanation*

On CPU, a bare DeiT-Tiny forward pass had a median latency of 49.6 ms, and a prediction that also returned a relevancy-propagation explanation cost 164.1 ms, about 3.3 times the bare-prediction cost. Both numbers use the same protocol (single-image batch, 5 warm-up iterations, median of 30), but the bare-forward and explanation pair (49.6 ms and 164.1 ms) was measured on the CPU of a cloud virtual machine during the abstention experiment, whereas Table 16 comes from the consumer laptop. The two figures are therefore not comparable in absolute terms; the 3.3 times ratio is a within-machine comparison.

4.7.2 *Synthetic degradation and OOD check*

EfficientViT-M0 degraded gradually, not catastrophically, under all three corruptions (Fig. 7). Gaussian blur was by far the most damaging, costing 32.8 points of balanced accuracy at its most severe level (radius 8), roughly double JPEG compression's worst case (15.5 points) and nearly double brightness/contrast's (18.1 points). A plausible reason is that Bethesda-system diagnosis depends on fine nuclear texture, which blur removes directly. On the 25-image synthetic OOD sample, mean max-softmax confidence fell from 0.985 (real in-distribution images) to 0.743, a real drop but far above the 0.20 chance level for this five-class task, so confidence thresholding alone gives only a weak OOD signal.

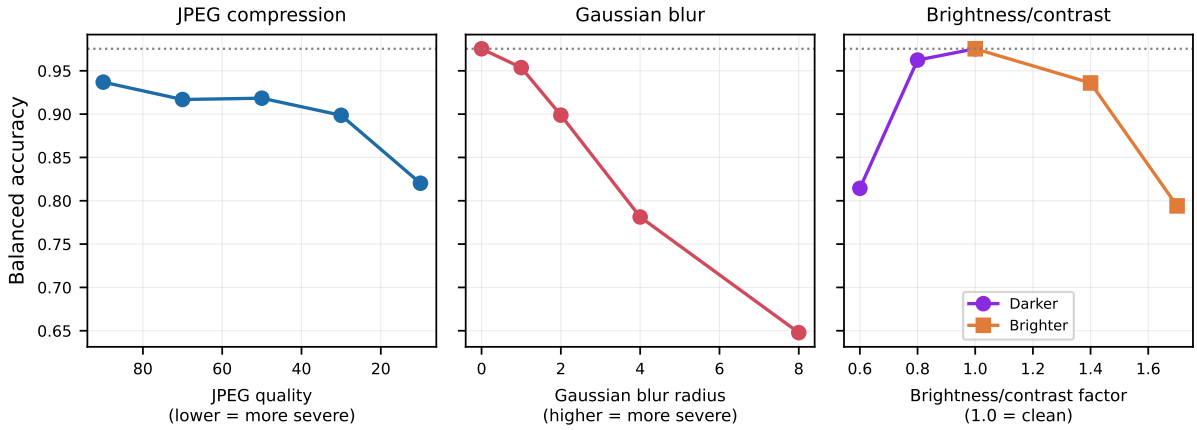


Figure 7: Balanced accuracy versus corruption severity for three synthetic degradations (EfficientViT-M0, SIPaKMeD test set). The dotted line marks the clean baseline (0.975). Gaussian blur causes the steepest decline.

5 Discussion

Read together, the results argue against several intuitive heuristics for selecting and verifying a deployment pipeline from a single evaluation. *"Smaller is worse"* is not supported in distribution (Sec. 4.1), although these in-distribution scores come from a split with substantial source-image overlap (a cluster-grouped re-run changed them only modestly, Sec. 4.4). *"The in-distribution leader is the safe choice"* is not supported: the leader ranked poorly under shift in the primary run, and even this ranking varies with the training seed (Sec. 4.3). *"Architecture lineage predicts robustness"* is contradicted by MobileViT-S and MobileViTv2, which share a lineage and produced a catastrophic failure and a top result on the same evaluation. *"CNNs are the robust family"* looked stable on Herlev and Mendeley LBC, but a third, independently sourced dataset (APCEM) did not support it, and a seed check showed that the Mendeley LBC split was itself seed-specific. Across the three datasets, no static property had an association with robustness that kept its direction (Table 8). The sharpest methodological point is that a robustness pattern that survives two external datasets is not guaranteed to survive a third, so "tested under two external shifts" is not a licence to stop testing.

These external results measure something narrower than deployment across institutions. Herlev, the dataset closest in content to the training data (single-cell crops), produced the highest external accuracies and a consistent pattern across seeds in which most transformer models outscored the CNNs. The two field-of-view datasets, Mendeley LBC and APCEM, ask a single-cell classifier to classify whole fields, and APCEM additionally changes the image appearance by enhancement. Below-chance results on those two datasets are therefore evidence that these classifiers do not transfer to field-level images under a normal/abnormal mapping, not evidence of how they would behave on single-cell crops from another institution. The robustness rankings should be read in that light.

The other three properties behaved differently from robustness and from each other. Explainability method choice produced the cleanest result, a single winner on every metric on both evaluated backbones, which suggests that choosing the method matters for these transformer backbones; the paired intervals on 596 cells support this, although concentration on the nucleus is not a clinical validity measure. Abstention composition was conditional on shift severity. Efficiency claims required direct measurement: FLOPs did not predict latency, an explanation cost about 3.3 times a bare prediction, and OOD confidence dropped only partially. No single verification strategy (pick the best method once, assume composition helps, trust an architectural efficiency claim) would have characterized all five properties correctly.

For deployment, the practical implications are direct. A backbone should not be chosen from an in-distribution leaderboard, however carefully measured, and robustness must be measured under

the shift the deployment will actually face, ideally across multiple seeds. Confidence-based abstention, even when combined with conformal set size, did not reach a reliable operating point under severe shift, so a screening tool should expect to route a substantial fraction of shifted-domain cases to specialist review. Blur is the most damaging of the tested corruptions, which makes focus quality in any handheld or low-cost capture setting a first priority for field validation.

6 Limitations

Data. All four datasets come from a small number of source institutions, and the three external shifts are real cross-dataset differences, not a systematic sweep of shift types (scanner resolution, stain intensity, or demographics varied independently). Because the datasets also differ in what an image contains (single cells versus fields of view), in preprocessing (APCEM enhancement), and in label definitions, the experiments do not separate institutional differences from scale, content, preprocessing, and annotation differences. The findings are demonstrated for twelve models and three external datasets; the observation that the tested static properties were not consistent predictors across datasets is the one most likely to generalize, whereas specific rankings, including which model is most robust, should not be assumed to transfer to a fourth dataset. Label harmonization to a binary task discards clinically important granularity and rests on category correspondences that the dataset documentation does not establish. SIPaKMeD (about 4,000 images) is a single-institution training source.

Source-image overlap. The in-distribution split is at the cell level, 87.8% of test cells share a source cluster image with a training cell, and no patient or slide identifiers are available (Sec. 3.3.1). In-distribution accuracy, calibration, abstention, and degradation results may therefore be optimistic. A cluster-level or patient-level re-split would be needed to estimate in-distribution performance on new patients, and it could change the in-distribution rankings reported here. The external-dataset results do not depend on this split. A single cluster-grouped re-run (Sec. 4.4) changed in-distribution balanced accuracy by a median of -0.9 points, comparable to seed variation, so it does not show a large inflation; however, that comparison is unpaired and uses one split, and a valid estimate needs a paired design (for example, repeated cell-level and cluster-grouped cross-validation on the same cells, with variability across which clusters fall in the test set). Earlier work has compared record-wise and subject-wise splitting on other medical datasets.[22,23]

Seeds, runs, and statistics. Each headline result is a single run at seed 13 on one data split. Seed variance was measured for all 12 models at two further seeds, and it showed that several single-model and single-dataset claims are seed-specific (the Mendeley LBC family split and EfficientViT-M0 as the in-distribution winner that fails under shift), so individual rows and close-ranked comparisons should be read with caution. Different tables come from separate runs of

unidentified run-to-run variation (Sec. 3.10), and same-seed repeatability was not measured. Per-image predictions and trained checkpoints were not retained for the original runs, so bootstrap confidence intervals, per-class sensitivity and specificity, and confusion matrices are available only for the cluster-grouped re-run; for the original runs we report seed statistics, rank correlations, and conservative sampling bounds (Sec. 3.11). The combined-signal abstention experiment was not repeated on the re-run. Reproducing a row means retraining from scratch, and exact software versions were not recorded (Table 3).

Backbone mismatch. The five properties were not all measured on one backbone: the benchmark and efficiency analysis cover 12 models, the explainability and abstention experiments use DeiT-Tiny (with ViT-Small for explainability), and the degradation and out-of-distribution experiments use EfficientViT-M0. DeiT-Tiny was used for explainability and abstention, not the benchmark's in-distribution leader, because relevancy propagation is not defined for EfficientViT-M0. The central observation therefore concerns differences across experiments, and we did not test whether evaluating all properties on one backbone would change it. The explainability ranking was confirmed on one additional same-family backbone (ViT-Small), but the abstention-composition result and the degradation and OOD findings were not re-tested on it. Whether any of these results transfer to a hybrid convolution–attention backbone, or to a different imaging task, is untested.

Measurement. The original explainability comparison used 50 cells and retained only means; the full-sample re-evaluation uses all 596 cluster-grouped test cells with paired intervals, but only for two related plain-transformer backbones, one deletion/insertion configuration (20 steps, blurred baseline), and one run. Deletion and insertion rankings can depend on the replacement baseline, and the nucleus concentration ratio measures overlap with an annotated nucleus, not clinical validity or reader agreement. The OOD check uses a small synthetic sample (25 images) rather than a natural out-of-domain dataset. CPU latency was measured on one shared consumer laptop (about 30% run-to-run swing) rather than dedicated or clinic-grade hardware, so the 3.3 times explanation-cost multiplier should be read as approximate. Accuracy, macro F1, and AUROC are standard metrics, which reduces the risk that the findings are artifacts of an unusual metric.

Clinical validity. No clinical or expert-review validation of explanation fidelity was performed. Post hoc explanations offer limited assurance about a model's reasoning unless they are validated,[58] and agreement with a nucleus boundary is not equivalent to a cytotechnologist judging an explanation clinically meaningful. No prospective or reader-study evaluation of the full pipeline was performed.

7 Conclusion

Five properties of single-cell cervical cytology classification, tested across experiments on a common dataset family and model registry, did not resolve alike. Parameter-efficient vision transformers did not trade accuracy for efficiency in distribution, yet the in-distribution leaderboard misled about robustness: none of the static properties we measured predicted robustness consistently across the three external datasets, and even apparent family-level patterns did not survive a third external dataset or a change of training seed. In-distribution scores came from a cell-level split with substantial source-image overlap and are likely optimistic. Relevancy propagation was more faithful than attention rollout or Grad-CAM on behavioral faithfulness and on expert-annotated localization on both evaluated transformer backbones. Combining two uncertainty signals for abstention helped only under the severe shift where the better individual signal was already failing. Efficiency claims held only under direct measurement, since FLOPs did not predict latency and an explanation cost roughly 3.3 times a bare prediction. Each property needed its own test, on its own terms, under the conditions of intended deployment.

Disclosures

The author declares that there are no financial interests, commercial affiliations, or other potential conflicts of interest that could have influenced the objectivity of this research or the writing of this paper.

Code, Data, and Materials Availability

All datasets analyzed are publicly available: SIPaKMeD,[48] Herlev,[49] Mendeley LBC,[50] and APCEM (`https://doi.org/10.17632/snzb5g6ddy.2`).[51] The training and evaluation code and the per-model result files are available from the corresponding author upon reasonable request.

Acknowledgments

This research received no external funding. Generative AI tools assisted with manuscript drafting and language refinement, under the author's direction, as described in the Materials and Methods. The author reviewed and verified the content and assumes full responsibility for the work.

References

1. H. Sung et al., "Global cancer statistics 2020: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries," *CA Cancer J. Clin.* **71**(3), 209–249 (2021) [doi:10.3322/caac.21660].
2. S. Mehta and M. Rastegari, "MobileViT: light-weight, general-purpose, and mobile-friendly vision transformer," in *Int. Conf. Learn. Represent. (ICLR)* (2022).
3. S. Mehta and M. Rastegari, "Separable self-attention for mobile vision transformers," arXiv preprint (2022).
4. H. Touvron, M. Cord, M. Douze, F. Massa, A. Sablayrolles, and H. Jégou, "Training data-efficient image transformers & distillation through attention," in *Int. Conf. Mach. Learn. (ICML)* (2021).
5. A. Dosovitskiy et al., "An image is worth 16x16 words: transformers for image recognition at scale," in *Int. Conf. Learn. Represent. (ICLR)* (2021).
6. B. Graham, A. El-Nouby, H. Touvron, P. Stock, A. Joulin, H. Jégou, and M. Douze, "LeViT: a vision transformer in ConvNet's clothing for faster inference," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 12239–12249 (2021) [doi:10.1109/ICCV48922.2021.01204].
7. K. Wu et al., "TinyViT: fast pretraining distillation for small vision transformers," in *Eur. Conf. Comput. Vis. (ECCV)* (2022).
8. X. Liu, H. Peng, N. Zheng, Y. Yang, H. Hu, and Y. Yuan, "EfficientViT: memory efficient vision transformer with cascaded group attention," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 14420–14430 (2023) [doi:10.1109/CVPR52729.2023.01386].
9. H. Cai, J. Li, M. Hu, C. Gan, and S. Han, "EfficientViT: lightweight multi-scale attention for high-resolution dense prediction," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 17256–17267 (2023) [doi:10.1109/ICCV51070.2023.01587].
10. M. Dehghani et al., "The efficiency misnomer," in *Int. Conf. Learn. Represent. (ICLR)* (2022), arXiv:2110.12894.
11. P. K. A. Vasu et al., "MobileOne: an improved one millisecond mobile backbone," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 7907–7917 (2023) [doi:10.1109/CVPR52729.2023.00764].
12. J. R. Zech et al., "Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs: a cross-sectional study," *PLoS Med.* **15**(11), e1002683 (2018) [doi:10.1371/journal.pmed.1002683].
13. H. Guan and M. Liu, "Domain adaptation for medical image analysis: a survey," *IEEE Trans. Biomed. Eng.* **69**(3), 1173–1185 (2022) [doi:10.1109/TBME.2021.3117407].
14. P. W. Koh et al., "WILDS: a benchmark of in-the-wild distribution shifts," in *Int. Conf. Mach.*

Learn. (ICML) (2021), arXiv:2012.07421.

15. R. Geirhos et al., "Shortcut learning in deep neural networks," *Nat. Mach. Intell.* **2**(11), 665–673 (2020) [doi:10.1038/s42256-020-00257-z].
16. A. D'Amour et al., "Underspecification presents challenges for credibility in modern machine learning," arXiv:2011.03395 (2020).
17. Y. Ovadia et al., "Can you trust your model's uncertainty? Evaluating predictive uncertainty under dataset shift," in *Adv. Neural Inf. Process. Syst. (NeurIPS)* (2019), arXiv:1906.02530.
18. S. C. B. Ferreira et al., "Assessing the efficacy of convolutional neural networks for Pap smear classification: a real world analysis," *PeerJ Comput. Sci.* **12**, e3708 (2026) [doi:10.7717/peerj-cs.3708].
19. G. Coskun, M. C. Akuner, and E. Kaplanoglu, "Robust cell-level classification for liquid-based cervical cytology using deep transfer learning: a multi-source study addressing scanner-induced domain shifts," *Bioengineering* **13**(3), 289 (2026) [doi:10.3390/bioengineering13030289].
20. N. Albzour and S. S. Lam, "Reliability-aware Hybrid-K ensemble selection for cervical cytology classification: integrating discrimination, calibration, and selective prediction," arXiv:2609.09189 (2026).
21. N. Bussola et al., "AI slipping on tiles: data leakage in digital pathology," in *Pattern Recognition. ICPR International Workshops and Challenges, Lecture Notes in Computer Science*, 167–182 (2021) [doi:10.1007/978-3-030-68763-2_13].
22. S. Saeb et al., "The need to approximate the use-case in clinical machine learning," *GigaScience* **6**(5), gix019 (2017) [doi:10.1093/gigascience/gix019].
23. D. J. Rumala, "How you split matters: data leakage and subject characteristics studies in longitudinal brain MRI analysis," in *Lecture Notes in Computer Science*, 235–245 (2023) [doi:10.1007/978-3-031-45249-9_23].
24. S. Abnar and W. Zuidema, "Quantifying attention flow in transformers," in *Proc. 58th Annu. Meet. Assoc. Comput. Linguist. (ACL)*, 4190–4197 (2020) [doi:10.18653/v1/2020.acl-main.385].
25. S. Jain and B. C. Wallace, "Attention is not explanation," in *Proc. 2019 Conf. North Am. Chapter Assoc. Comput. Linguist. (NAACL)*, 3543–3556 (2019) [doi:10.18653/v1/N19-1357].
26. S. Wiegreffe and Y. Pinter, "Attention is not not explanation," in *Proc. 2019 Conf. Empir. Methods Nat. Lang. Process. (EMNLP)*, 11–20 (2019) [doi:10.18653/v1/D19-1002].
27. H. Chefer, S. Gur, and L. Wolf, "Transformer interpretability beyond attention visualization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 782–791 (2021) [doi:10.1109/CVPR46437.2021.00084].
28. R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM:

visual explanations from deep networks via gradient-based localization," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, 618–626 (2017) [doi:10.1109/ICCV.2017.74].

29. V. Petsiuk, A. Das, and K. Saenko, "RISE: randomized input sampling for explanation of black-box models," in *Br. Mach. Vis. Conf. (BMVC)* (2018).
30. L. Barekatain and B. Glocker, "Evaluating the explainability of vision transformers in medical imaging," in *Lecture Notes in Computer Science*, 96–105 (2026) [doi:10.1007/978-3-032-17611-0_10].
31. M. Chung et al., "Evaluating visual explanations of attention maps for transformer-based medical imaging," in *Lecture Notes in Computer Science*, 110–120 (2025) [doi:10.1007/978-3-031-77610-6_11].
32. P. Komorowski, H. Baniecki, and P. Biecek, "Towards evaluating explanations of vision transformers for medical imaging," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, 3726–3732 (2023) [doi:10.1109/CVPRW59228.2023.00383].
33. C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Int. Conf. Mach. Learn. (ICML)* (2017).
34. A. N. Angelopoulos and S. Bates, "Conformal prediction: a gentle introduction," *Found. Trends Mach. Learn.* **16**(4), 494–591 (2023) [doi:10.1561/2200000101].
35. Y. Gal and Z. Ghahramani, "Dropout as a Bayesian approximation: representing model uncertainty in deep learning," in *Int. Conf. Mach. Learn. (ICML)* (2016).
36. B. Lakshminarayanan, A. Pritzel, and C. Blundell, "Simple and scalable predictive uncertainty estimation using deep ensembles," in *Adv. Neural Inf. Process. Syst. (NeurIPS)* (2017).
37. M. Sensoy, L. Kaplan, and M. Kandemir, "Evidential deep learning to quantify classification uncertainty," in *Adv. Neural Inf. Process. Syst. (NeurIPS)* (2018).
38. M. T. Hagos et al., "Validation of conformal prediction in cervical atypia classification," *Sci. Rep.* **16**(1), 9649 (2026) [doi:10.1038/s41598-026-44850-5].
39. N. Albzour and S. S. Lam, "Selective prediction and uncertainty-aware referral for Pap smear classification," arXiv:2609.17545 (2026).
40. W. Liu et al., "Is the aspect ratio of cells important in deep learning? A robust comparison of deep learning methods for multi-scale cytopathology cell image classification: from convolutional neural networks to visual transformers," arXiv:2105.07402 (2021).
41. M. V. Lima, M. H. M. Britto, and L. Britto Neto, "Classificação de células cervicais cancerígenas com CNNs e ViTs em ensemble para auxílio ao diagnóstico médico," in *Anais do XXV Simpósio Brasileiro de Computação Aplicada à Saúde (SBCAS)*, 497–508 (2025) [doi:10.5753/sbcas.2025.7467].
42. N. Albzour and S. S. Lam, "Systematic evaluation of vision transformers for automated cervical cancer classification: optimization, statistical validation, and clinical interpretability,"

Cancers **18**(13), 2178 (2026) [doi:10.3390/cancers18132178].

43. M. M. Rahaman et al., "DeepCervix: a deep learning-based framework for the classification of cervical cells using hybrid deep feature fusion techniques," *Comput. Biol. Med.* **136**, 104649 (2021) [doi:10.1016/j.compbiomed.2021.104649].
44. A. B. Mohiuddin, M. M. Rahman, M. N. Gony, S. M. K. Shuvra, A. Rafy, M. R. Ahmed, and S. Rahman, "Accelerated and accurate cervical cancer diagnosis using a novel stacking ensemble method with explainable AI," in *2025 Int. Conf. Quantum Photonics, Artif. Intell., Networking (QPAIN)*, 1–6 (2025) [doi:10.1109/QPAIN66474.2025.11171850].
45. M. M. Rahman et al., "MaizeFormerX: a lightweight vision transformer with cross-scale attention for explainable maize leaf disease diagnosis," *Sci. Rep.* **16**, 15160 (2026) [doi:10.1038/s41598-026-44550-0].
46. M. N. Gony, S. M. K. Shuvra, R. Poudel, K. Dhakal, M. M. Rahman, A. Rafy, M. R. Ahmed, and F. M. Alamgir, "Dual-view semi-supervised retinal disease recognition via chromatic structural decomposition and cross-view contrastive learning," in *2026 Int. Conf. Power, Electron., Commun., Comput., Intell. Infrastruct. (PECCII)*, 1–6 (2026) [doi:10.1109/PECCII70991.2026.11661813].
47. M. Ahmed, F. Ahmed, S. M. Mow, P. A. Taha, S. Barua, M. M. Rahman, A. Rafy, S. M. Mondol, and M. I. Faisal, "An explainable multimodal AI framework with reinforcement learning for post-surgical clinical decision support," medRxiv 2026.06.08.26355217 (2026).
48. M. E. Plissiti, P. Dimitrakopoulos, G. Sfikas, C. Nikou, O. Krikoni, and A. Charchanti, "SIPAKMED: a new dataset for feature and image based classification of normal and pathological cervical cells in Pap smear images," in *2018 25th IEEE Int. Conf. Image Process. (ICIP)*, 3144–3148 (2018) [doi:10.1109/ICIP.2018.8451588].
49. J. Jantzen, J. Norup, G. Dounias, and B. Bjerregaard, "Pap-smear benchmark data for pattern classification," in *Proc. Nature Inspired Smart Inf. Syst. (NiSIS)*, 1–9 (2005).
50. E. Hussain, L. B. Mahanta, H. Borah, and C. R. Das, "Liquid based-cytology Pap smear dataset for automated multi-class diagnosis of pre-cancerous and cervical cancer lesions," *Data Brief* **30**, 105589 (2020) [doi:10.1016/j.dib.2020.105589].
51. H. Aswini, "APCEM-Enhanced Cervical Cytology Image Dataset, Version 2," Mendeley Data (2026) [doi:10.17632/snzb5g6ddy.2].
52. K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 770–778 (2016) [doi:10.1109/CVPR.2016.90].
53. M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "MobileNetV2: inverted residuals and linear bottlenecks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 4510–4520 (2018) [doi:10.1109/CVPR.2018.00474].

54. M. Tan and Q. Le, "EfficientNet: rethinking model scaling for convolutional neural networks," in *Int. Conf. Mach. Learn. (ICML)* (2019).
55. Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo, "Swin Transformer: hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 9992–10002 (2021) [doi:10.1109/ICCV48922.2021.00986].
56. D. Picard, "Torch.manual_seed(3407) is all you need: on the influence of random seeds in deep learning architectures for computer vision," arXiv:2109.08203 (2021).
57. X. Bouthillier et al., "Accounting for variance in machine learning benchmarks," in *Mach. Learn. Syst. (MLSys)* (2021), arXiv:2103.03098.
58. M. Ghassemi, L. Oakden-Rayner, and A. L. Beam, "The false hope of current approaches to explainable artificial intelligence in health care," *Lancet Digit. Health* **3**(11), e745–e750 (2021) [doi:10.1016/S2589-7500(21)00208-9].